



BAULKHAM HILLS HIGH SCHOOL

2015

YEAR 12 HALF-YEARLY

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions
- Start a new page for each question

Total marks – 70

Exam consists of 8 pages.

This paper consists of TWO sections.

Section 1 – Pages 2-4

Multiple Choice

Question 1-10 (10 marks)

Section 2 – Pages 5-8

Extended Response

Question 11- 14 (60 marks)

Section I

10 marks

Attempt Questions 1-10

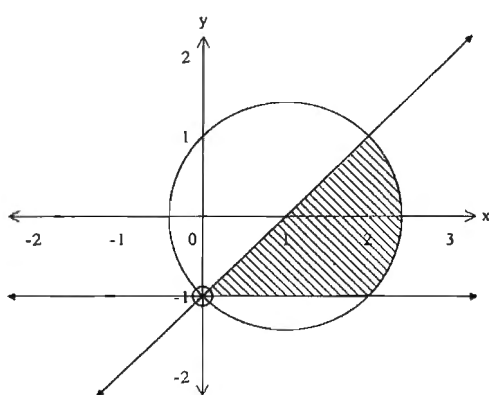
Allow about 15 minutes for this section

Use the multiple choice answer sheet in your booklet for Questions 1-10

1. What is the remainder when $x^3 + x^2 + 5x + 6$ is divided by $x + i$?

- (A) $5 - 4i$
- (B) $5 + 6i$
- (C) $7 - 6i$
- (D) $7 - 4i$

2. Consider the Argand diagram below.

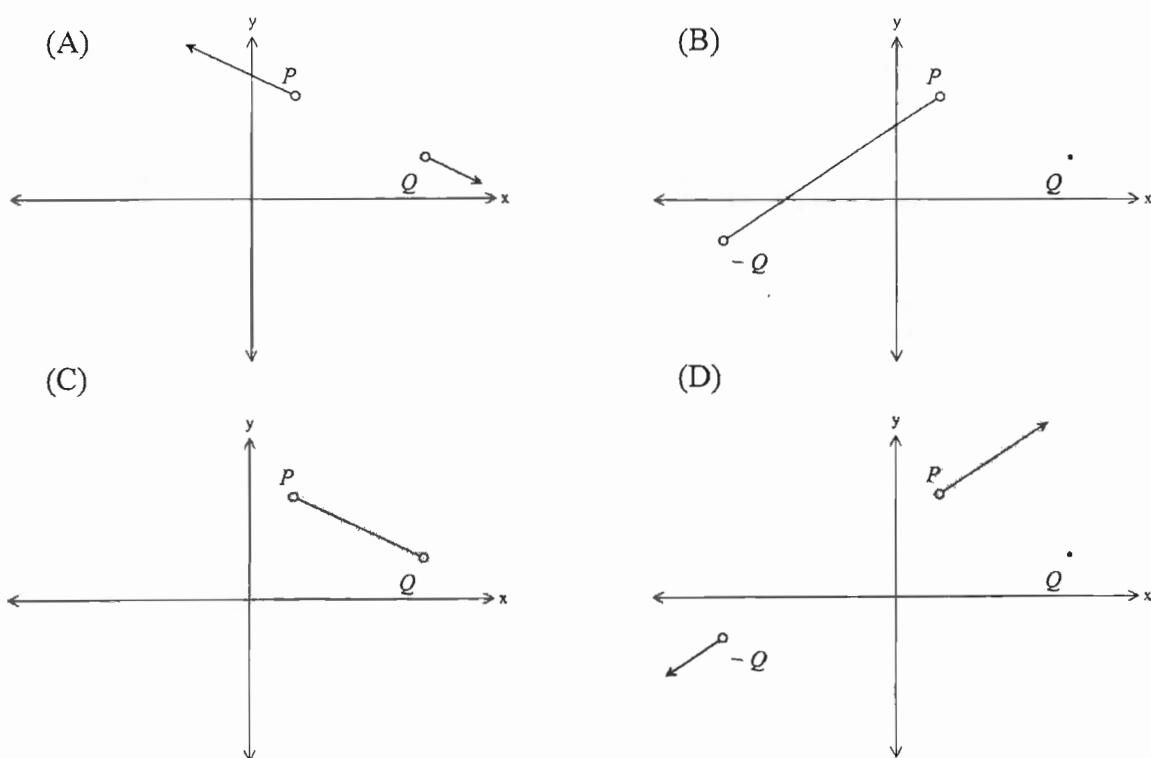


Which pair of inequalities define the shaded area?

- (A) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
 - (B) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$
 - (C) $|z - 1| \leq 1$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
 - (D) $|z - 1| \leq 1$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$
3. Given $3x + 2iy - ix + 5y = 7 + 5i$, where x and y are real numbers, then:
- (A) $x = -1, y = 2$
 - (B) $x = \frac{39}{11}, y = -\frac{8}{11}$
 - (C) $x = -\frac{3}{5}, y = \frac{22}{5}$
 - (D) $x = -11, y = 8$

4. If z represents a variable point on the Argand diagram, which description best represents the locus of $|z - 4 + i| = |z + 4 - i|$?
- (A) Circle
(B) Ellipse
(C) Hyperbola
(D) Line
5. If ω is a non-real cube root of unity the value of $\frac{1}{1 + \omega} + \frac{1}{1 + \omega^2}$ is equal to
- (A) -1
(B) 0
(C) 1
(D) ω
6. The polynomial equation $P(z) = 0$ has real coefficients, and has roots including a double root of $-3 + 2i$ and a single root of 3.
What is the minimum possible degree of $P(z)$?
- (E) 3
(F) 4
(G) 5
(H) 6
7. The points P , Q and R are represented by the complex numbers p , q and r respectively.
The points P , Q and R are joined to form a triangle.
If $r + pi = q + ri$ what is the value of the ratio of $RQ:PQ$?
- (A) $\frac{1}{\sqrt{2}}$
(B) 1
(C) $\sqrt{2}$
(D) 2

8. P and Q are two points representing the complex numbers p and q respectively.
Which of the following shows the locus of z defined by $\arg(z + q) - \arg(z - p) = \pi$?



9. If $\log_b a = c$ and $\log_x b = c$, then $\log_a x$ equals

- (A) a
(B) c^{-2}
(C) b
(D) b^2

10. For what values of k will $\frac{x^2}{|2k| - 3} + \frac{y^2}{k - 2} = 1$ be a hyperbola?

- (A) $-\frac{3}{2} < k < \frac{3}{2}$ or $k < 2$
(B) $\frac{3}{2} < k < 2$
(C) $k < 2$
(D) $k < -\frac{3}{2}$ or $\frac{3}{2} < k < 2$

End of Section I

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question on the relevant page of your writing booklet.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Answer on the relevant page of your writing booklet.

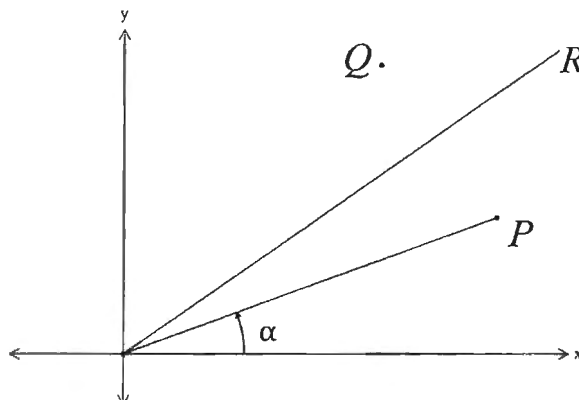
- (a) Consider the complex number $z = \sqrt{3} + i$
- (i) Express z in modulus-argument form. 1
 - (ii) Write down $\frac{1}{z^2}$ in modulus-argument form. 1
 - (iii) Find the least positive value of n for which z^n is a positive real number. 1
- (b) Sketch the locus of z if $\arg\left(\frac{z-3}{z+5}\right) = \frac{2\pi}{3}$ 2
- (c) Find integers m and n such that $(x+1)^2$ is a factor of $x^5 + 2x^2 + mx + n$ 3
- (d) Consider the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.
- (i) Find the eccentricity 1
 - (ii) Find the equation of the directrices 1
 - (iii) Find the coordinates of the foci. 1
 - (iv) Sketch the graph of the ellipse, clearly showing all intercepts with the axes and the foci and directrices. 1
- (e) The product of two of the roots of the cubic $6x^3 - 23x^2 + kx - 12 = 0$ is 2.
- (i) Find the value of k . 1
 - (ii) Find all the roots of the cubic equation. 2

End of Question 11

Question 12 (15 marks) Answer on the relevant page of your writing booklet.

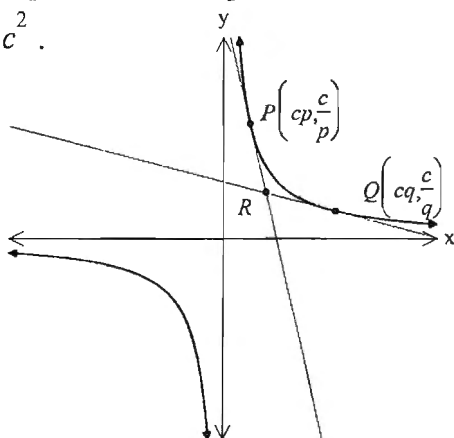
- (a) Let $P(z) = z^2 + 8 + 6i$
- (i) Solve $P(z) = 0$ 2
- (ii) Hence, or otherwise solve $z^2 - (5 - i)z + 8 - i = 0$ 2

- (b) The point P represents the complex number z . The reflection of P in the line OR , making an angle of α with the positive direction of the real axis, is the point Q , representing the complex number w .



- (i) Explain why $|z| = |w|$ 1
- (ii) Hence prove that $zw = |z|^2 (\cos 2\alpha + i \sin 2\alpha)$ 2

- (c) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are two distinct points on the rectangular hyperbola $xy = c^2$.

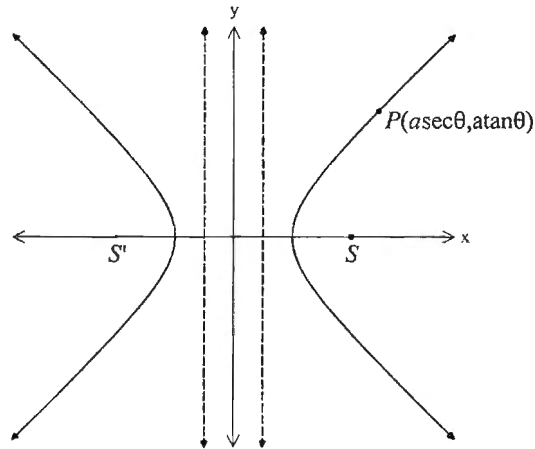


- (i) Prove that the equation of the tangent at P is $x + p^2 y = 2cp$. 2
- (ii) The tangents at P and Q meet at R . Find the coordinates of R . 2
- (iii) The midpoint of PQ is M . Find the coordinates of M . 2
- (iv) If O is the origin, prove that O, R and M are collinear. 2

End of question 12

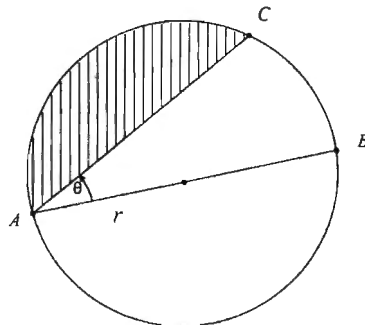
Question 13 (15 marks) Answer on the relevant page of your writing booklet.

- (a) Solve $\frac{x^2 + x - 6}{x^2 - 4x} \leq 1$ 3
- (b) The roots of the cubic equation $x^3 - 6x^2 - 11 = 0$ are α , β and γ . Form the equation which has roots $\alpha + \beta$, $\beta + \gamma$, and $\gamma + \alpha$. 3
- (c) The point $P(a \sec \theta, a \tan \theta)$ is a point on the rectangular hyperbola $x^2 - y^2 = a^2$.



The foci are S and S' .

- (i) Show that the eccentricity is $\sqrt{2}$. 1
- (ii) Show that $SP = a\sqrt{2} \left(\sec \theta - \frac{1}{\sqrt{2}} \right)$. 2
- (iii) Hence show that $SP \cdot S'P = OP^2$, where O is the origin. 2
- (d) In the diagram, AB is the diameter of the circle (radius r) and $\angle BAC = \theta$.



- (i) If the area of the shaded region is one third of the area enclosed by the circle, show that $\sin 2\theta = \frac{\pi}{3} - 2\theta$. 2
- (ii) By drawing appropriate graph/s, show that the above equation has only one solution. 2

End of Question 13

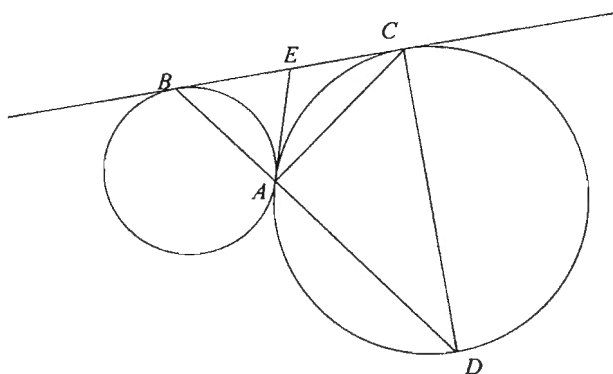
Question 14 Start on the appropriate page in your answer booklet

(a) Given $\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$ (DO NOT PROVE THIS RESULT)

(i) Find the roots of $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ 2

(ii) Hence, without the aid of a calculator, prove that 3
 $\sec^2 \frac{\pi}{16} + \sec^2 \frac{3\pi}{16} + \sec^2 \frac{5\pi}{16} + \sec^2 \frac{7\pi}{16} = 32$

(b) Two circles touch at A . BC and AE are common tangents. BA is produced to meet the second circle at D .



(i) Prove that $AC \perp AB$ 2

(ii) Prove that the lengths AB , AC and AD are in geometric progression. 2

(c) (i) Sketch the curve $|z - \alpha| = |\alpha|$ where α is a complex number. 2

(ii) Hence find the maximum value of $|z - 2\text{Re}(\alpha)|$ 1

(d) (i) In how many ways is it possible to separate 5 distinct objects into two different boxes so that no box remains empty? 1

(ii) In how many ways is it possible to factor the number 30 030 into three positive integer factors greater than 1 (note: $30 \times 77 \times 13$ is the same factoring as $13 \times 30 \times 77$). 2

End of Paper

Section I - Multiple choice

1. A $P(-i) = -i^2 + i^2 - 5i + 6$
 $= 5 - 4i$

$\therefore A$

2. B circle has centre $(1, 0)$ radius $= \sqrt{4+1}$
 $= \sqrt{5}$

$$\arg(z - (-i)) = \arg(z + i)$$

$\therefore B$

3. A Equating real and imaginary parts

$$3x + 5y = 7 \quad (1)$$

$$2y - x = 5 \quad (2)$$

sub (2) in (1) $3(2y - 5) + 5y = 7$
 $6y - 15 + 5y = 7$
 $11y = 22$

$$y = 2$$

$$x = 2 \times 2 - 5$$

$$x = -1$$

$\therefore A$

4. D Locus is of form $|z - z_1| = |z - z_2|$ which results in perpendicular bisector of interval z_1 to z_2 .

$\therefore D$

5. C $\frac{1}{1+w} + \frac{1}{1+w^2} = -\frac{1}{w^2} + \frac{-1}{w}$ using $1+w+w^2=0$

$$= \frac{-1-w}{w^2}$$

$$= \frac{w^2}{w^2}$$

$$= 1$$

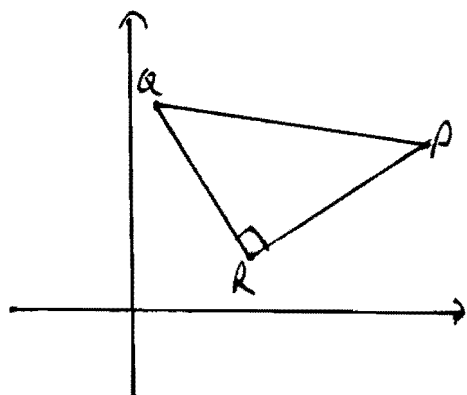
$\therefore C$

6. C $P(z) = (z+3-2i)^2 (z-3) (z+3-2i)^2 Q(z)$

$\begin{matrix} \uparrow \\ -3+2i \\ \text{is double} \\ \text{root} \end{matrix}$
 $\begin{matrix} \uparrow \\ z=3 \text{ is} \\ \text{single root} \end{matrix}$
 $\begin{matrix} \uparrow \\ \text{coefficients are} \\ \text{real, roots occur} \\ \text{in conjugate pairs} \end{matrix}$

$\therefore \text{degree} = 2+2+1 + \deg(Q(z))$
 $\therefore \text{minimum degree} = 5$ (if $Q(z)$ is a constant)
 $\therefore C$

7. A



$r+pi = q+tri$
 $i(p-r) = q-r$
 $\therefore \vec{PR} = \vec{QR}$
 $\therefore |PR| = |QR|$ & $|PQ| = \sqrt{2}$
 $\therefore \frac{RQ}{PQ} = \frac{1}{\sqrt{2}}$

$\therefore A$

8. B Section between P and -Q
 $\therefore B$

since $\arg(z+q) - \arg(z-p)$
 $= \arg(b+q) - \arg(z-p)$

9. B $b^c = a$ & $x^c = b$

$(x^c)^c = b^c$
 $x^{c^2} = b^c$
 $x^{c^2} = a$
 $(x^c)^{\frac{1}{c^2}} = a^{\frac{1}{c^2}}$
 $\log_a x = c^{-2}$
 $\therefore B$

10. Either $|2k|-3 > 0$ & $k-2 < 0$ OR $|2k|-3 < 0$ & $k-2 > 0$

$|2k| > 3$ $|2k| < 3$ & $k > 2$
 $\therefore k < -\frac{3}{2}$ or $k > \frac{3}{2}$ & $k < 2$ $-\frac{3}{2} < k < \frac{3}{2}$ & $k > 2$
 $\therefore \text{No sol'n}$

$\therefore \text{Solution is } k < -\frac{3}{2} \text{ or } \frac{3}{2} < k < 2$
 $\therefore D$

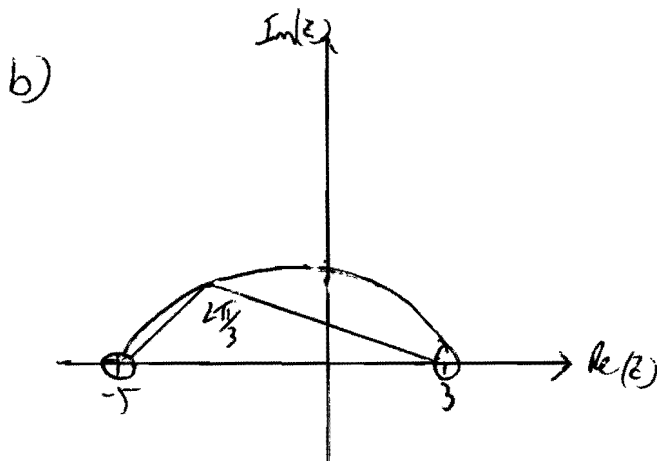
SECTION II

Q 11 a) (i) $z = 2 \cos\left(\frac{\pi}{6}\right)$ ✓

(ii) $z^{-2} = 2^{-2} \cos\left(\frac{\pi}{6}\right)^{-2}$
 $\frac{1}{z^2} = \frac{1}{4} \cos\left(\frac{-\pi}{3}\right)$ ✓

(iii) $z^n = 2^n \cos\left(\frac{n\pi}{6}\right)$
 $\therefore$ for z^n to be positive real number $\frac{n\pi}{6}$ must be a multiple of 2π
 $\therefore \frac{n\pi}{6} = 2\pi k$ for k , an integer

$\therefore n = 12$



(2) minor arc or $\frac{2\pi}{3}$ from $k = -5$
 (1) arc from allow (1) for arc

c) $x = -1$ is a double root.
 $P(-1) = 0 = -1 + 2 - m + n$
 $P'(-1) = 0 = 5 - 4 + m$

$\therefore m = -1$

$\therefore 2 + n = 0$

$n = -2$

$\therefore n = -1$ & $n = -2$.

$P(x) = x^5 + 2x^2 + mx + n$
 $P'(x) = 5x^4 + 4x + m$

(3) correct

(2) correct m

(1) using $P'(x) = 0$

d) i) $b^2 = a^2(1 - e^2)$
 $4 = 9(1 - e^2)$
 $\frac{4}{9} = 1 - e^2$
 $e^2 = \frac{5}{9}$
 $e = \frac{\sqrt{5}}{3}$ ✓

$$(3x-4)(x-3)=0$$

∴ Roots are $1, \frac{4}{3}, \frac{3}{2}$. ✓

12 a) i)

$$z^2 = -8-6i$$

Let $z = x+iy$
 $x^2 - y^2 + 2ixy = -8-6i$

$$x^2 - y^2 = -8$$

$$2xy = 6$$

$$xy = 3$$

$$x = \pm 3, y = \pm 1$$

$$\therefore z = \pm (3-i)$$

(ii)

$$z = \frac{5-i \pm \sqrt{(5-i)^2 - 4(8-i)}}{2}$$

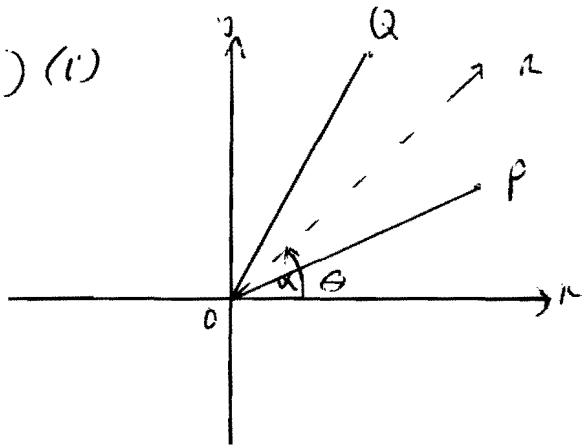
$$z = \frac{5-i \pm \sqrt{24-10i-32+4i}}{2}$$

$$= \frac{5-i \pm \sqrt{-8-6i}}{2}$$

$$= \frac{5-i \pm (1-3i)}{2}$$

$$z = 3-2i \text{ or } 2+i$$

b) (i)



Since Q is a reflection of P in OR, $PQ \perp OR$ and midpoint of PQ is on OR.

∴ $\triangle OPQ$ is isosceles.

$$\therefore OP = OQ$$

(i) correct reasoning

b(ii) let arg $P = \theta$

$$\angle POR = \angle -\theta$$

Also $\angle POR = \angle ROQ$ (Q is reflection of P in OR)

$$\therefore \arg q = \alpha + (\alpha - \theta)$$

$$= 2\alpha - \theta$$

$$zw = |z||w| \operatorname{cis} (\theta + 2\alpha - \theta)$$

$$= |z|^2 \operatorname{cis} 2\alpha$$

✓ since $|w| = |z|$

c) i) $y = \frac{c^2}{x}$

$$\frac{dy}{dx} = -c^2 x^{-2}$$

$$= -\frac{c^2}{x^2}$$

At P , $\frac{dy}{dx} = \frac{-c^2}{c^2 p^2}$

$$\frac{dy}{dx} = -\frac{1}{p^2}$$

② correct solution

① correct $\frac{dy}{dx}$

Eqn of tangent $y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp \quad \text{①} \checkmark$$

ii) Tangent at Q is $ktq^2 y = 2cq \quad \text{②}$

for R : ① - ② $(p^2 - q^2)y = 2c(p - q)$

$$(p+q)(p-q)y = 2c(p-q)$$

$$y = \frac{2c}{p+q}$$

$$\therefore x + p^2 \left(\frac{2c}{p+q} \right) = 2cp$$

$$x = 2cp - \frac{2p^2 c}{p+q}$$

$$= \frac{2cp(p+q) - 2p^2 c}{p+q}$$

$$x = \frac{2cpq}{p+q}$$

② correct solution

① either x or y value correct

$$\therefore R \text{ is } \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right) \checkmark$$

$$(iii) M = \left(\frac{cp+cq}{2}, \frac{\frac{c}{p} + \frac{c}{q}}{2} \right)$$

$$M = \left(\frac{c(p+q)}{2}, \frac{c(p+q)}{2pq} \right) \checkmark$$

$\uparrow$ (1) $\uparrow$ (1)

(2) correct solution
 (1) either x or y value correct

$$(iv) \text{ For } R \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right) \Rightarrow x = \frac{2cpq}{p+q} \quad y = \frac{2c}{p+q}$$

$$\therefore x = pqy$$

$$\text{i.e. } y = \frac{x}{pq} \quad \therefore R \text{ lies on line gradient } \frac{1}{pq} \text{ through } (0,0) \checkmark$$

$$\text{For } M, \quad x = \frac{c(p+q)}{2} \quad y = \frac{c(p+q)}{2pq}$$

$$\therefore y = \frac{x}{pq}$$

which also has same gradient $\frac{1}{pq}$ as OR and passes through $(0,0)$ ✓

$\therefore O, R, M$ are collinear ✓

OR { either (1) for gradient of OR or OM
 (2) for correct solution

(a) bottom of next page

b) $x^3 - 6x^2 - 11 = 0$ has roots α, β, γ
 $\alpha + \beta + \gamma = 6$

New roots: $\alpha + \beta = 6 - \gamma$
 $\beta + \gamma = 6 - \alpha$
 $\gamma + \alpha = 6 - \beta$ } $\therefore$ we want roots $y = 6 - x$ where $x = \alpha, \beta, \gamma$

$\therefore (6-x)^3 - 6(6-x)^2 - 11 = 0$
 $216 - 3 \times 6^2 \times x + 3 \times 6x^2 - x^3 - 6(36 - 12x + x^2) - 11 = 0$
 $-x^3 + 12x^2 - 36x - 11 = 0$
 $\therefore x^3 - 12x^2 + 36x + 11 = 0$

c) (i) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
 $b^2 = a^2(e^2 - 1)$
 $1 = e^2 - 1$
 $e^2 = 2$
 $e = \sqrt{2}$

(ii) Direction $x = \pm \frac{a}{e}$
 $x = \pm \frac{a}{\sqrt{2}}$

$L = \frac{PS}{1m}$

$PS = \sqrt{2} PM$

$= \sqrt{2} \left(a \sec \theta - \frac{a}{\sqrt{2}} \right)$

$= a\sqrt{2} \left(\sec \theta - \frac{1}{\sqrt{2}} \right)$

Similarly replacing a with $\frac{a}{\sqrt{2}}$ yields $PS' = a\sqrt{2} \left(\sec \theta + \frac{1}{\sqrt{2}} \right)$

(iii)

$SP \cdot SP' = a\sqrt{2} \left(\sec \theta - \frac{1}{\sqrt{2}} \right) \left(a\sqrt{2} \left(\sec \theta + \frac{1}{\sqrt{2}} \right) \right)$
 $= 2a^2 \left(\sec^2 \theta - \frac{1}{2} \right)$

$OP^2 = (a \sec \theta)^2 + (a \tan \theta)^2$
 $= a^2 (\sec^2 \theta + \tan^2 \theta)$

$= a^2 (\sec^2 \theta + \sec^2 \theta - 1)$

$= a^2 (2\sec^2 \theta - 1)$

$= 2a^2 \left(\sec^2 \theta - \frac{1}{2} \right)$

$= SP \cdot SP'$ as reqd.

- 13 d) (i) $\triangle AOC$ is isosceles (equal radii)
 $\therefore \angle ACO = \theta$
 $\therefore \angle AOC = \pi - 2\theta$ (L sum of $\triangle AOC$)

Area segment = Area of Circle

$$\frac{1}{2} r^2 \left[(\pi - 2\theta) - \sin(\pi - 2\theta) \right] = \frac{\pi r^2}{3}$$

✓ ① expression for area of segment with circle

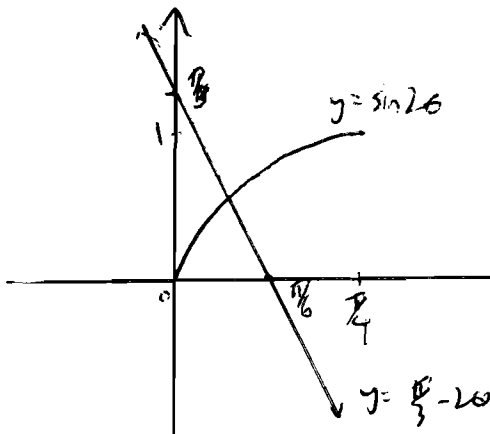
$$\pi - 2\theta - \sin 2\theta = \frac{2\pi}{3}$$

$$\sin 2\theta = \pi - \frac{2\pi}{3} - 2\theta$$

$$\sin 2\theta = \frac{\pi}{3} - 2\theta$$

✓

ii)



② correct solution

① one correct graph drawn with points indicated

$\therefore$ one point of intersection is one root

13(a) $\frac{x^2 + x - 6}{x^2 - 4x} \leq 1$
 $\frac{(x+3)(x-2)}{x(x-4)} \leq 1 \quad x \neq 0, 4$

Critical points:

$$\frac{(x+3)(x-2)}{x(x-4)} = 1$$

$$x^2 + x - 6 = x^2 - 4x$$

$$5x = 6$$

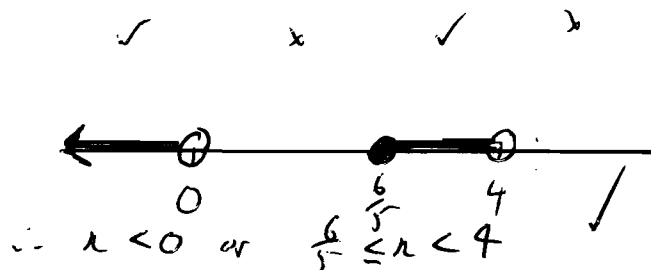
$$x = \frac{6}{5}$$

③ correct solution

② 1 correct region

① critical values/multiply by $(x^2 - 4x)^2$

or



14 a) i) let $\tan 4\theta = 1$ & let $x = \tan \theta$

alternative solutions
at end

$$1 = \frac{4x - 4x^3}{1 - 6x^2 + x^4}$$

$$x^4 - 6x^2 + 1 = 4x - 4x^3$$

$x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$ has roots given by $x = \tan \theta$ ✓
where $\tan 4\theta = 1$

$$\therefore 4\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$$

$$\therefore \theta = \frac{\pi}{16}, \frac{5\pi}{16}, \frac{9\pi}{16}, \frac{13\pi}{16}$$
 ✓

$$\therefore \text{roots are } x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$$

(or $x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{7\pi}{16}, -\tan \frac{3\pi}{16}$)

(2i)

$$\text{Eqn of roots } \tan^2 \frac{\pi}{16}, \tan^2 \frac{5\pi}{16}, \tan^2 \frac{9\pi}{16}, \tan^2 \frac{13\pi}{16}$$

will be found by letting $y = x^2$
 $\therefore x = \sqrt{y}$

$$y^2 + 4y\sqrt{y} - 6y - 4\sqrt{y} + 1 = 0$$

$$(y^2 - 6y + 1)(y^2 - 4\sqrt{y} + 1) = [4\sqrt{y}(1-y)]^2$$

$$y^4 - 6y^3 + y^2 - 4y^2\sqrt{y} + 4y\sqrt{y} - 4\sqrt{y} + 1 = 16y(1-2y+y^2)$$

$$y^4 - 28y^3 + 70y^2 - 28y + 1 = 0$$
 ✓

$$\therefore \tan^2 \frac{\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{9\pi}{16} + \tan^2 \frac{13\pi}{16} = 28$$

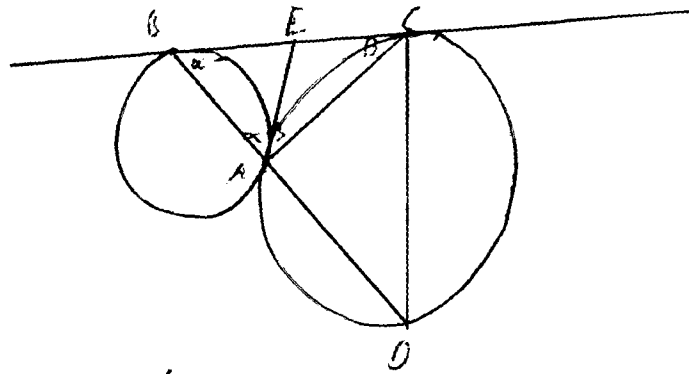
$$\text{But } \tan^2 \theta = \sec^2 \theta - 1$$

$$\therefore \sec^2 \frac{\pi}{16} + \sec^2 \frac{5\pi}{16} + \sec^2 \frac{9\pi}{16} + \sec^2 \frac{13\pi}{16} - 4 = 28$$

$$= 32 \quad \left. \begin{array}{l} \text{✓} \text{ found} \\ \text{as reqd.} \end{array} \right\}$$

See end for alternative method & marking scheme.

b)



(i) tangent at A meet BC at E

let $\angle EBA = \alpha$

$\angle EAB = \alpha$ (tangents from external point are equal, $\angle EAB$ is $\angle EBA$) ✓

similarly let $\angle ECA = \beta$
 $\angle EAC = \beta$

Now In $\triangle BAC$

$\angle B + \angle C + \angle A = 180^\circ$ ($\angle$ sum of $\triangle BAC$)

$2(\alpha + \beta) = 180^\circ$

$\alpha + \beta = 90^\circ$

$\therefore \angle CAE = 90^\circ$

$\therefore AC \perp AB$ ✓

ii) In $\triangle BAC$ & $\triangle ACD$

$\angle BCA = \angle CDA = \beta$ (alternate segment theorem)

$\angle CAB = \angle CAD = 90^\circ$ ($AC \perp AB$ in (i))

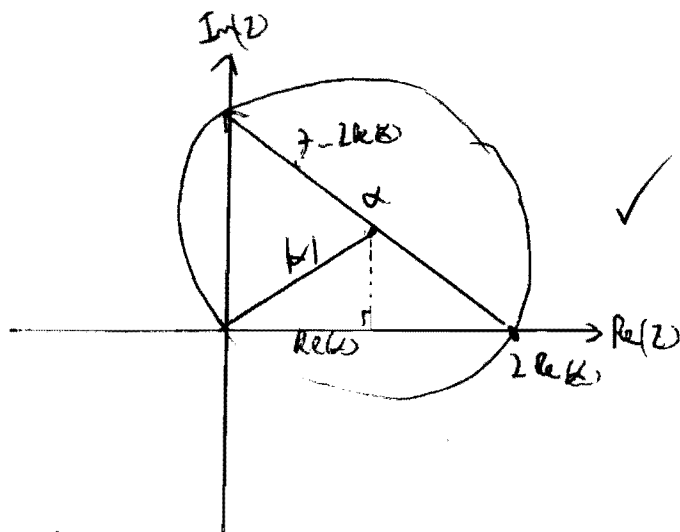
$\therefore \triangle ABC \sim \triangle ACD$ (AA)

$\therefore \frac{CA}{AB} = \frac{AB}{AC}$ (matching sides in same ratio, of similar $\triangle$ s) ✓

$\therefore \frac{CA}{AB} = \frac{AD}{AC}$

$\therefore AB, AC, AD$ are in geometric progression ✓

c) (i)



(2) correct
(1) Circle centre α (not on real axis)

ii)

For max value, $|z - 2\text{Re}(\alpha)|$, z must lie on diameter on circle radius α .

$$\therefore |z - 2\text{Re}(\alpha)| = 2|\alpha| \quad \checkmark$$

d(i) $2^5 - 2$
 $= 30$

ii) Factors: 2, 3, 5, 7, 11, 13

for any method:

(2) correct
(1) significant progress

$$\begin{aligned} & \underbrace{\quad}_{\text{factor 1}} \times \underbrace{\quad}_{\text{factor 2}} \times \underbrace{\quad}_{\text{factor 3}} \\ &= \frac{3^6 - 3 \times 2^6 + 1}{3!} \\ &= \frac{540}{6} \\ &= 90 \end{aligned}$$

(each factor has 3 choices but we need to subtract no of ways of having one empty box 3×2^6 AND add back in all three empty)

Alternatively using (a)

$$\begin{aligned} & \text{Possible combinations} - 1 \text{ box empty} - 2 \text{ boxes empty} \\ &= \frac{3^6 - 3(2^6 - 2) - 3 \times 1^6}{3!} \\ &= \frac{540}{6} \\ &= 90 \end{aligned}$$